

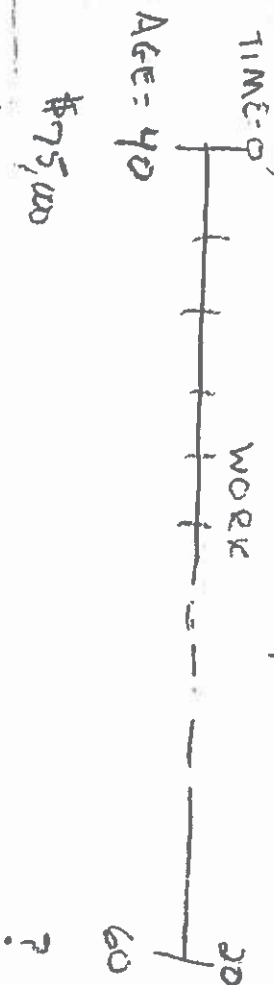
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Assume that your father is now 40 years old, that he plans to retire in 20 years, and that he expects to live for 25 years after he retires, that is, until he is 85. He wants a fixed retirement income that has the same purchasing power at the time he retires as \$75,000 has today (he realizes that the real value of his retirement income will decline year-by-year after he retires). His retirement income will begin the day he retires, 20 years from today, and he will then get 2% additional annual payment. Inflation is expected to be 4 percent per year from today forward; he currently has \$200,000 saved up; and he expects to earn a return on his savings of 7 percent per year, annual compounding. To the nearest dollar, how much must he save during each of the next 20 years (with deposits being made at the end of each year) to meet his retirement goal?

#5 FATHER

1. Father saves for 20 yrs, then receives pmts for 25 yrs
2. Wants payments of \$75,000/yr in today's \$ for 1st pmt only. Real income declines, Inflation of 4%.

So, find inflated fixed pmts:

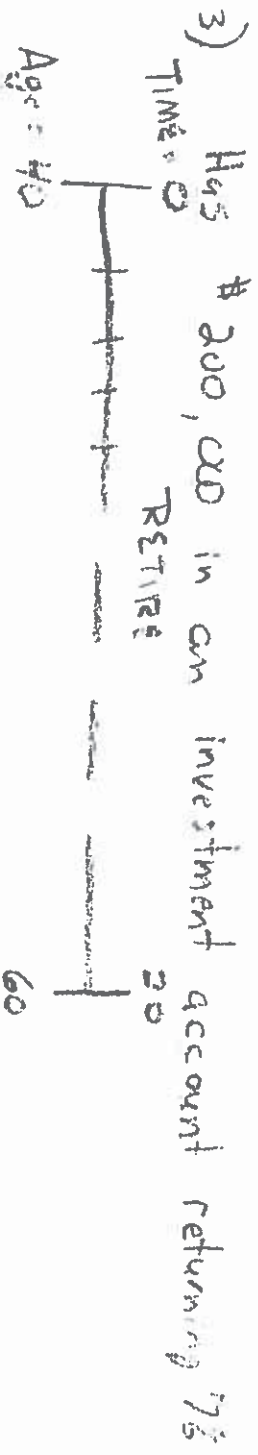


$$PV < 75,000 >$$

$$N = 20$$

$$(\text{inflation}) i = 4\%$$

$$FV = ? = 164,334$$



$$PV = < 200,000 >$$

$$N = 20$$

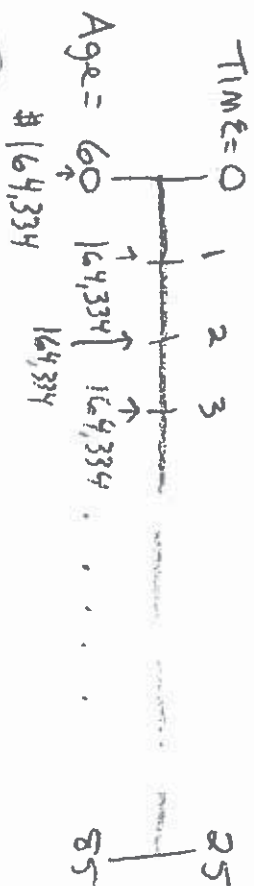
$$i = 7\%$$

$$PMT = 0$$

$$FV = ? = \$177,937$$

25 years

4. Wants retirement payments (after adjusted for inflation) of \$164,334 annually made at the beginning of the first retirement year. So, how much will he need accumulated to be able to receive these annual payments for 25 years?



BEC

$$N = 25$$

$$i = 7$$

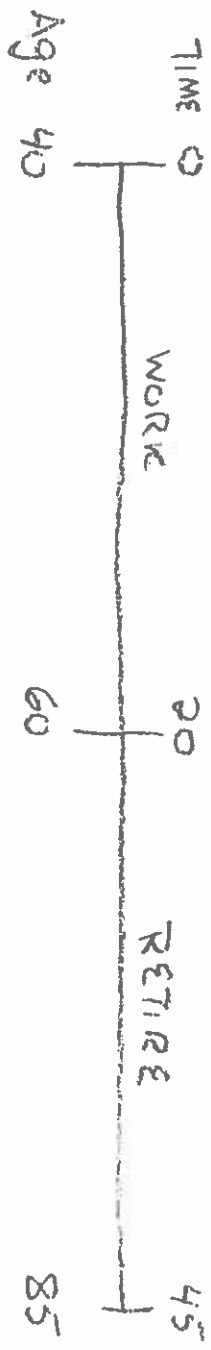
$$FV = 0$$

$$PMT = 164,334$$

$$PV = ? = \$2,049,138$$

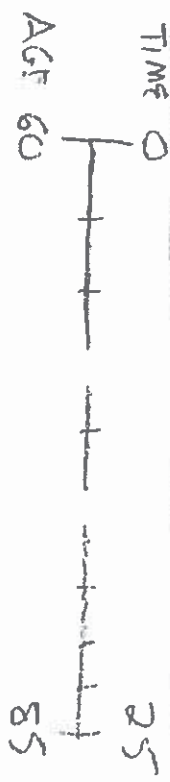
45 value

5) Original \$200,000 @ age 40 grows to \$773,937 @ age 60



\$200,000

\$773,937

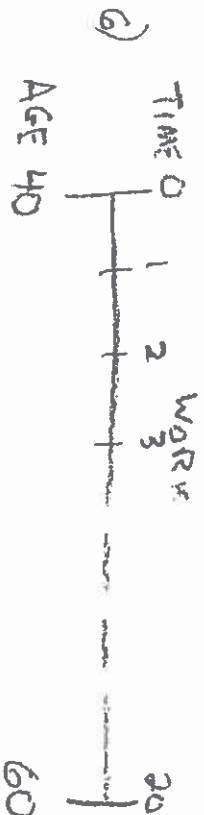


Need PV of

Annuity = \$2049,135

Have = \$773,937

Add'l amt = \$1,275,201



Needs

\$1,275,201 $\Rightarrow$ So, how much needs to

be invested yearly during working years?

FV = 1,275,201

N = 20

i = 7%

PV = 0

PMT = ? = 31,106